

Conjecture 00000002190: disproof

Conjecture. “The Sperner capacity of C_{2k+1} equals $\cos(\pi/(2k+1))$.”

Reading. C_{2k+1} is the cycle on $\text{Fin } (2k+1)$, with vertex i adjacent to $i+1 \bmod (2k+1)$. The conjecture is for all $k \geq 1$; we refute it at $k = 1$, where $\cos(\pi/3) = 1/2$. For a digraph with arc relation E a *Sperner code* of length n is a set S of words in V^n such that for every ordered pair $x \neq y$ in S there is a coordinate i with $E(x_i, y_i)$. M_n is the largest size of such a code. “Capacity = c in unit u ” means $u(M_n^{1/n}) \rightarrow c$, with $u = \text{id}$ (exponential units), $u = \log_2$ (bits) or $u = \ln$ (nats). An orientation of C_{2k+1} has arcs only along cycle edges and exactly one direction on each edge $\{i, i+1\}$ (Lean: `IsOrientation`); the conjecture is read as “for some orientation”. We also treat the undirected zero-error (Shannon) capacity: a code in which distinct words have a coordinate with distinct non-adjacent letters. Conventions: capacity is a limit; the proof bounds the $\liminf$ from below, so $\limsup$ and $\sup$ readings fail as well. Not refuted: base-4 logarithm at $k = 1$ for the cyclic triangle (capacity 2 would give exactly $1/2$), and nat units at individual $k \geq 2$ (not needed).

Proof

Sperner. Any orientation has an arc $a \rightarrow b$, $a \neq b$ (Lean: `xor` at vertex 0). For $T \subseteq [n]$ let w_T be b on T and a elsewhere. If $|T| = |T'| = \lfloor n/2 \rfloor$ and $T \neq T'$, then $T' \not\subseteq T$, so some $j \in T' \setminus T$ has $(w_T(j), w_{T'}(j)) = (a, b)$. Hence $M_n \geq \binom{n}{\lfloor n/2 \rfloor}$, and since $\sum_i \binom{n}{i} = 2^n$ with $n+1$ terms, none exceeding the middle one, $2^n \leq (n+1)M_n$ (`arc_bound`). So $M_n^{1/n} \geq 2/(2^{1/n}n^{1/n}) \rightarrow 2$ (`g_le`, `g_tendsto`), using $n+1 \leq 2n$. For a unit u monotone on $(0, \infty)$ and continuous at 2, a limit c satisfies $c \geq u(2)$ (`reading_ge`). Now $u(2) = 2, 1, \ln 2 > 0.69$ for `id`, `log2`, `ln`, all exceeding $\cos(\pi/3) = 1/2$ (`sperner_refuted`).

Shannon. In $C_3 = K_3$ any two distinct letters are adjacent, so a code has at most one word and $M_n = 1$ (`shannon_M`). Then $u(M_n^{1/n}) = u(1) \in \{1, 0, 0\} \neq 1/2$ (`shannon_refuted`).

Lean

File `lean/Conjecture2190/Basic.lean` (namespace `C2190`); statement block between the `STATEMENT` markers; main theorem `C2190.main : Claim`, where `Claim` is the conjunction of `¬ConjSperner` and `¬ConjShannon` in the three units. `#print axioms C2190.main`: `propext`, `Classical.choice`, `Quot.sound`. No sorry, no `native_decide`.